



American International University-Bangladesh (AIUB)

Department of Computer Science

Faculty of Science & Technology (FST)

Project Title:
Hospital Management System

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Submitted by

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1. PROJECT PROPOSAL

1.1 Background to the Problem

Hospitals are complex organizations where many departments work together to provide proper patient care. Doctors, nurses, receptionists, laboratory staff, and patients all depend on accurate information and smooth communication. In many hospitals, especially small and medium-sized ones, patient records, appointment schedules, admission details, and test reports are still handled manually or through separate systems. This traditional way of management creates several problems such as data duplication, loss of records, delays in service, and human errors. Managing large amounts of patient data using paper files also makes the process slow and inefficient.

Patients often face difficulties in booking appointments, getting admitted, or collecting test reports on time. Doctors may not have quick access to patient history, and nurses may find it hard to track the current condition of admitted patients. Similarly, receptionists and laboratory staff need constant coordination with other departments, which becomes difficult without a centralized system. These issues not only waste time but also affect the quality of healthcare services.

To overcome these problems, a computerized Hospital Management System is required. This system brings all hospital activities under one platform where each user can access the system based on their role. Patients can book doctor appointments and view test reports, doctors can manage patient visits and assign tests, nurses can update patient conditions, receptionists can handle admissions and doctor information, and phlebotomists can submit test reports. Overall, the system helps improve efficiency, accuracy, data security, and communication within the hospital.

1.2 Selection of Process Model

For the development of this Hospital Management System, the **Scrum model**, which is an Agile software development approach, has been selected. Scrum is suitable for this project because it allows the system to be developed in small and manageable parts instead of building everything at once. The project is divided into short development cycles called sprints, where each sprint focuses on a specific set of features such as user authentication, appointment booking, admission management, or test reporting.

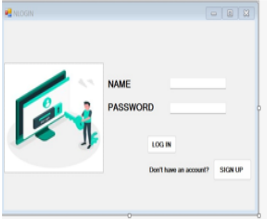
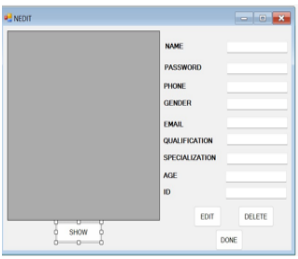
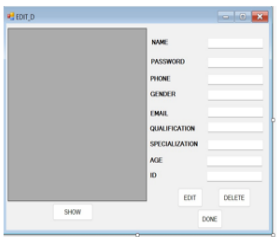
One of the main advantages of the Scrum model is flexibility. Requirements can be reviewed and improved after each sprint based on feedback from team members or instructors. This is especially helpful for an academic project, where understanding of requirements may improve during development. Scrum also encourages regular communication through meetings such as sprint planning, daily stand-ups, sprint review, and sprint retrospective, which helps the team stay organized and focused.

Another reason for choosing Scrum is that it supports teamwork and continuous improvement. Each team member has clear responsibilities, and progress can be monitored easily after every sprint. Testing is also done regularly, which helps identify errors early and improve system quality. Due to its iterative nature, ease of management, and adaptability, the Scrum model is an appropriate and effective choice for developing the Hospital Management System in this software engineering course.

2. SOFTWARE REQUIREMENTS SPECIFICATIONS (SRS) / PRODUCT REQUIREMENTS DOCUMENT (PRD)

2.1 Scopes and Features

A	B	C	D	E	F	G	H
Product Requirement Document (PRD) - Version: 1.0							
Template Name		Hospital Management System.					
Goals & Objective		The main goal of this Hospital Management System is to automate and streamline hospital operations by providing a unified, user-friendly application. It will improve how receptionists, doctors, nurses, and patients interact with hospital records. This system increases efficiency, reduces manual workload, and ensures secure and accurate data handling, ultimately improving patient care and coordination among hospital staff. -Centralize and securely store all patient, doctor, nurse, and receptionist data in a SQL Server database. -Provide secure, role-based login and registration for patients, doctors, nurses, and receptionists. -Enable receptionists to admit patients, schedule doctor appointments, and assign doctors efficiently. -Allow doctors and nurses to update patients' health status, test recommendations, and admission or follow-up orders in real time.					
Product Type		Web Application					
Domain		Web2					
Access Layers		Patient, doctor, Nurse, Receptionist					
Design Required		Yes					
Phase	Item No	Specification [Role]	Screen Definition	User Story	Acceptance Criteria	UI Screen / Wireframe	Tags
Feature: 1 Auth Feature							
phase 1	1	Signup [Patient] [doctor][Nurse][Receptionist]	<Form> <input=phone> <input=gender> <input=email> <input=address> <input=qualification> <input=specialization> <input=experience> <input=blood group> <input=age> <button=signup> <link=signin>		Phone input must contain a valid phone number (Should be numeric and 11 digits (Bangladesh standard). If invalid, show: "Please enter a valid phone number." Gender must be selected User must choose Male/Female/Other. If not selected, show: "Please select your gender." Email input must contain a valid, unused email address Must follow a valid email format. If email already exists in the database, show: "Email already used." Address field cannot be empty If left blank, show: "Address is required." Qualification must be provided Cannot be empty. If empty, show: "Please enter your qualification." Specialization is required for doctors only If user type = Doctor and field is empty, show: "Please enter your specialization." Experience field must contain a number Only numeric values accepted. If invalid, show: "Please enter valid experience (in years)." Blood group input must be a valid type Accept only: A+, A-, B+, B-, AB+, AB-, O+, O- If invalid, show: "Invalid blood group." Age must be a number between 1-120 If empty or invalid, show: "Please enter a valid age." All required fields must be filled If any required field is empty, show: "Please fill all the required fields."		signup
phase 1	2	Login [Patient] [doctor][Nurse][Receptionist]	<Form> <input=name> <input=password> <button=login> <link=signup>		- Name must match database email - password must match database password - User go to dashboard (if name & password match with database) - User is prompt with an error. "Name or password does not match". (if name & password match with database) - clicking signup should take user to signup page		login
Feature: 2 Dashboard							
phase 1	3	Menu [Patient]	Dashboard Book Doctor View Profile Edit Profile Change Password		- clicking dashboard will take user to dashboard page - go to book doctor on Click - go to view profile page on click - go to edit profile page on Click - go to change password page on Click		dashboard
phase 1	3	Menu [Nurse]	Dashboard Admitted Patients View Profile Edit Profile Change Password		- clicking dashboard will take user to dashboard page - go to admitted patients on Click - go to view profile page on click - go to edit profile page on Click - go to change password page on Click		dashboard

14	phase 1	2	Login[Patient] [doctor][Nurse][Receptionist]	<Form> <input=name> <input=password> <button=login> <link=signup>		- Name must match database email - password must match database password - User go to dashboard (if name & password match with database) - User is prompt with an error. "Name or password does not match". (if name & password match with database) - clicking signup should take user to signup page		login
15	Feature: 2 Dashboard							
16	phase 1	3	Menu [Patient]	Dashboard Book Doctor View Profile Edit Profile Change Password		- clicking dashboard will take user to dashboard page - go to book doctor on Click - go to view profile page on click - go to edit profile page on Click - go to change password page on Click		menu
17	phase 1	3	Menu [Nurse]	Dashboard Admitted Patients View Profile Edit Profile Change Password		- clicking dashboard will take user to dashboard page - go to admitted patients on Click - go to view profile page on click - go to edit profile page on Click - go to change password page on Click		menu
22	A	B	C	D	E	F	G	H
23		4	Admit patient[receptionist]	<table> <input= name> <input= cause> <input= recommendation> <input= room no> <input= bed no> <input= condition> <button=book>		- No null values - valid room no and bed no should given		admit
24	phase 1	5	View Profile[Patient] [doctor][Nurse][Receptionist]	<table> <input = Name > <input = Password > <input = Phone > <input =Email > <input =Address> <radio button = Gender > <radio button = Blood Group> <input =Email > <input = Qualification > <input = Specialization> <input =Expreience> <date input = Age > <date input = Id > <button = Show> <button = Edit> <button = Delete> <button = Done>		- No null values - Edit button should take user to edit page		view
25	phase 1	6	Edit Profile[Patient] [doctor][Nurse][Receptionist]	<table> <input = Name > <input = Password > <input = Phone > <input =Email > <input =Address> <radio button = Gender > <radio button = Blood Group> <input =Email > <input = Qualification > <input = Specialization> <input =Expreience> <date input = Age > <date input = Id > <button = Show> <button = Edit> <button = Delete> <button = Done>		- No null values - email should be in correct format [xyz@abc.com] - phone number should be in correct format [11 digits] - date of birth should be in at least 15 years - clicking save button will either prompt with an error if above rules do not meet or a prompt notifying the user about successful change		change

2.2 User Story Table

as a/an	i want to	so that	acceptance criteria
Patient	Create a new patient profile	I can enroll myself into the system	Valid name and phone number
	login with username and password	i can use the system functions	data should match to signup
	view my profile	i can see the list of information given in the time of registration	information list with edit profile option
	edit my profile	i can update the incorrect information	information list window with edit button
	view doctor list	book the doctor as i want	clicking the doctor list
	appoint doctor	visit the doctor in time	give valid information and timing
Doctor	Create a new doctor profile	I can enroll myself into the system	Valid name ,email and phone number
	login with username and password	I can use the system functions	Username and password must match the information given during signup
	View my profile	I can see the list of information given in the time of registration	information list should load with profile details
	Edit my profile	I can update the incorrect information	Information list window with edit information button and updates must save correctly
	View appointments	I can see all my scheduled and past appointments clearly	Appointment list should show patient name, date, time, and status

	Update appointments	I can manage and correct appointment details when needed	Doctor can update appointment status (confirmed/cancelled)
Nurse	Login with my username and password	The system can authenticate me and I can trust it	After successful login, all data associated to me should be accessible
	View my own profile details	I can see the information provided during my registration	Profile table should display all stored information. Edit profile button should be visible
	Update my information	I can correct mistakes or update my personal data	No field should be left empty. Edit window must save changes and show a confirmation message
	Change my password	My account can be more secure	Must enter old password. Must enter new password twice. System sends a success notification
	Request a new password	I can regain access if I forget my password	OTP is sent to user's mobile. Confirmation email is also sent
	View admitted patients	I can monitor patients under hospital care	List of all admitted patients must be shown. Patient details should be correctly mapped
	Update patient status	I can keep doctors and staff informed	Status dropdown should update database. Success message should appear after saving
	Manage medicine inventory (view only or limited access)*	I can assist in delivering correct medicine during rounds	System shows updated medicine list. Search bar functions properly
Receptionist	Create a new receptionist profile	I can enroll myself into the system	Valid name and phone number

	login with username and password	i can use the system functions	data should match to signup
	View my profile	i can see the list of information given in the time of registration	information list window with edit button
	Edit my profile	i can update the incorrect information	information list window with edit button
	Admit patient	can register the patient into the hospital for treatment	available bed or ward must be selected available bed or ward must be selected
	Make Schedule	I can arrange appointments or doctor timings properly	valid schedule information must be saved and displayed correctly
Phlebotomist	Create a new phlebotomist profile	I can enroll myself into the system	Valid name and phone number
	login with username and password	i can use the system functions	data should match to signup
	View my profile	i can see the list of information given in the time of registration	information list window with edit button
	Edit my profile	i can update the incorrect information	information list window with edit button
	Submit report	i can submit patient test report	Valid patient test details must be entered before submission
	Booked Test List	i can book patient test list	<i>Booked test list window should display all scheduled patient tests</i>

2.3 Requirements Traceability Matrix

The Requirements Traceability Matrix (RTM) is used to map system requirements with their corresponding system modules and test cases. It ensures that every requirement defined for the Hospital Management System is properly designed, implemented, and tested. RTM helps in tracking requirements throughout the software development life cycle and confirms that no requirement is left unaddressed.

2.3.1 Functional Requirements

Req. ID	Requirement Description	User Type	System Module	Test Case ID
FR-01	User can sign up into the system	All Users	Authentication Module	TC-01
FR-02	User can log in and log out	All Users	Authentication Module	TC-02
FR-03	User can update personal information	All Users	Profile Management	TC-03
FR-04	Patient can view doctor list and timings	Patient	Doctor Information Module	TC-04
FR-05	Patient can book doctor appointment	Patient	Appointment Management	TC-05
FR-06	Patient can view test reports	Patient	Report Management	TC-06
FR-07	Receptionist can add/update doctor information	Receptionist	Doctor Management	TC-07
FR-08	Receptionist can admit patients	Receptionist	Admission Management	TC-08
FR-09	Doctor can view patient booking list	Doctor	Appointment Management	TC-09

FR-10	Doctor can update patient condition	Doctor	Patient Record Module	TC-10
FR-11	Doctor can assign medical tests	Doctor	Test Assignment Module	TC-11
FR-12	Nurse can view admitted patient list	Nurse	Admission Management	TC-12
FR-13	Nurse can update patient condition	Nurse	Patient Record Module	TC-13
FR-14	Phlebotomist can view assigned test list	Phlebotomist	Test Management	TC-14
FR-15	Phlebotomist can submit test reports	Phlebotomist	Report Management	TC-15

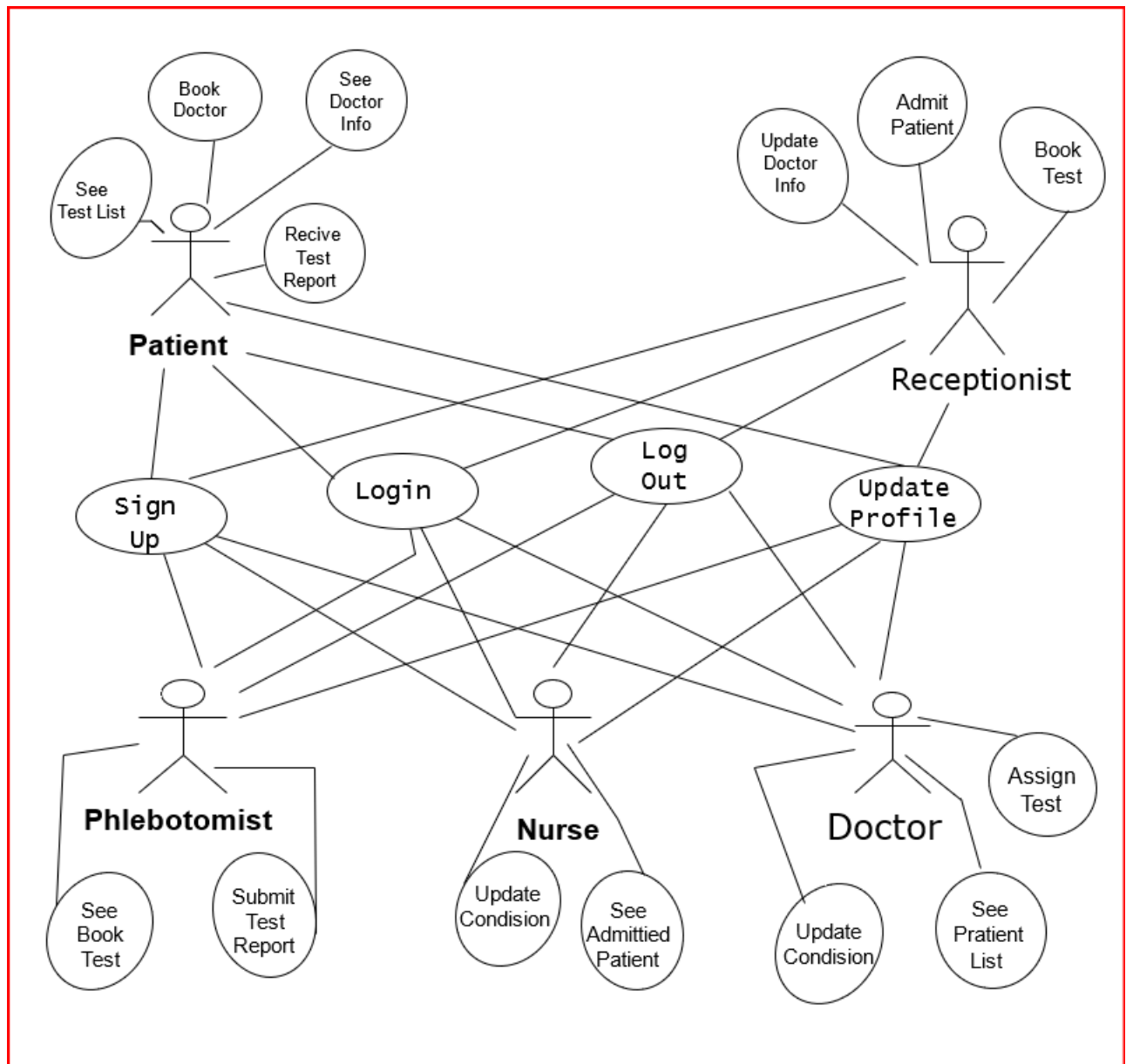
2.3.2 Non-Functional Requirements

Req. ID	Non-Functional Requirement	Description	Validation Method
NFR-01	Usability	System should be easy to use for all users	User Testing
NFR-02	Performance	System should respond quickly to user actions	Performance Testing
NFR-03	Security	User data must be protected using authentication	Security Testing
NFR-04	Reliability	System should work without failure during use	System Testing
NFR-05	Availability	System should be accessible during hospital hours	Operational Testing
NFR-06	Scalability	System should support future feature expansion	Design Review
NFR-07	Data Integrity	Stored data should remain accurate and consistent	Database Testing

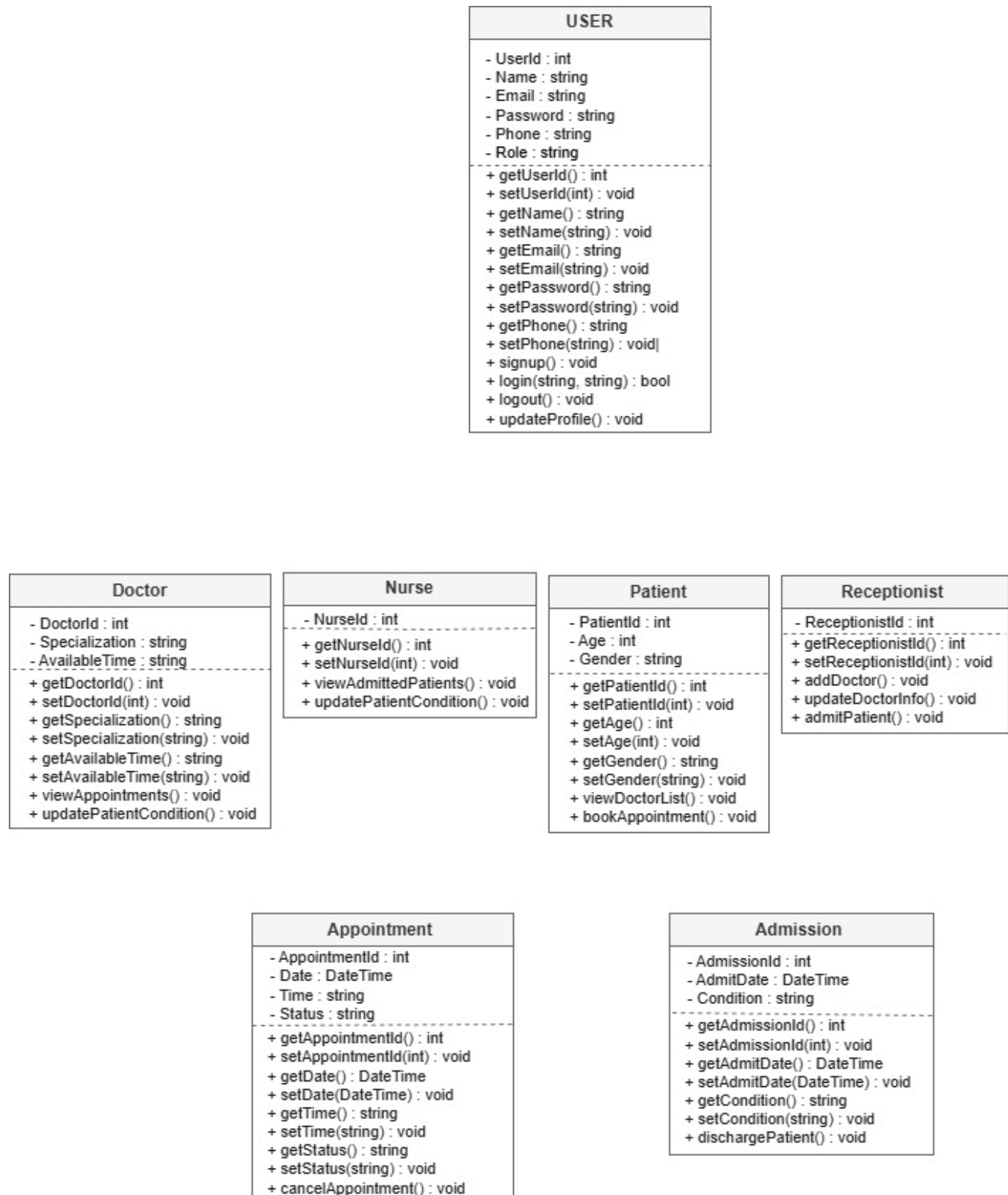
3. SOFTWARE DESIGN

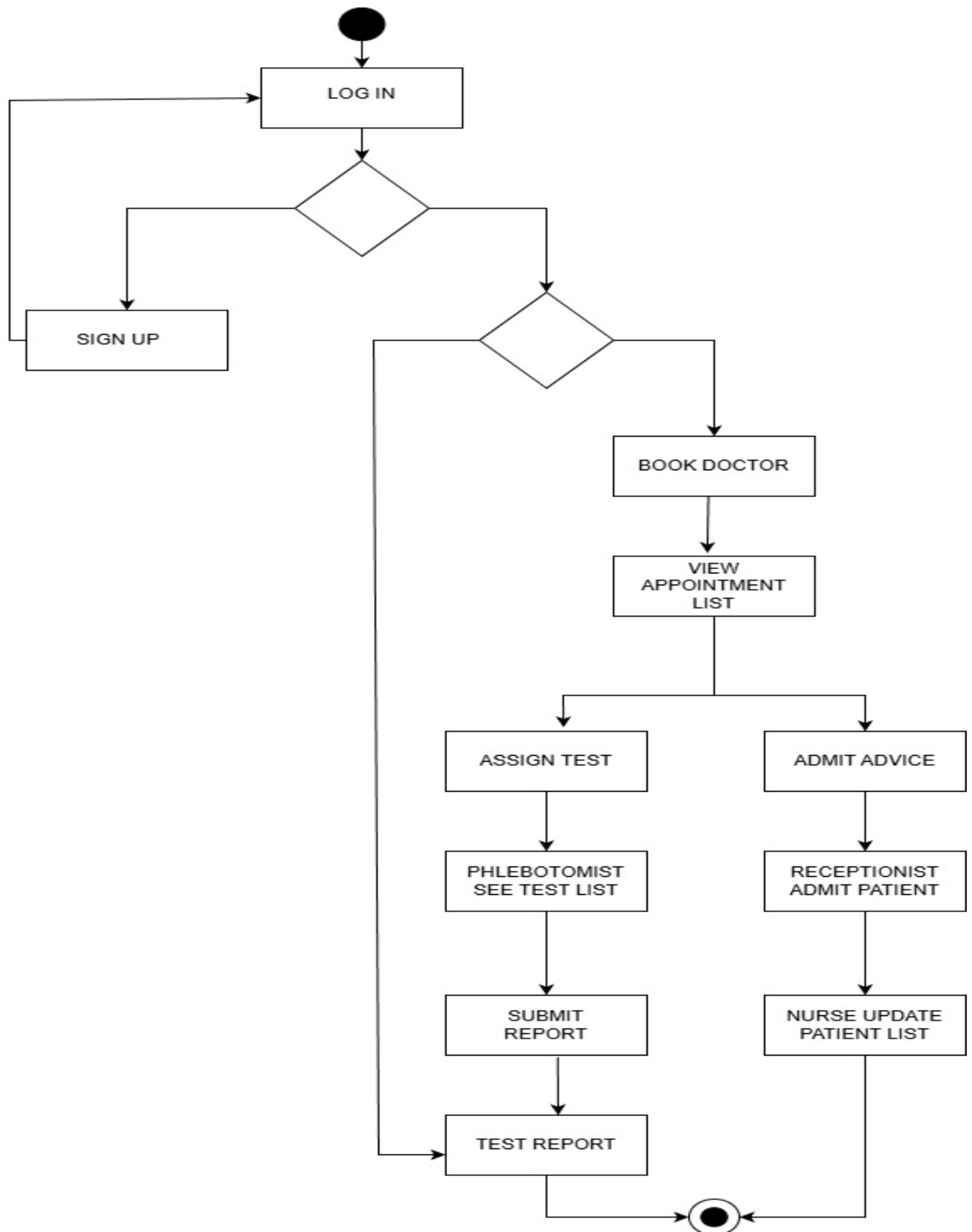
3.1 System Design

Use Case Diagram:



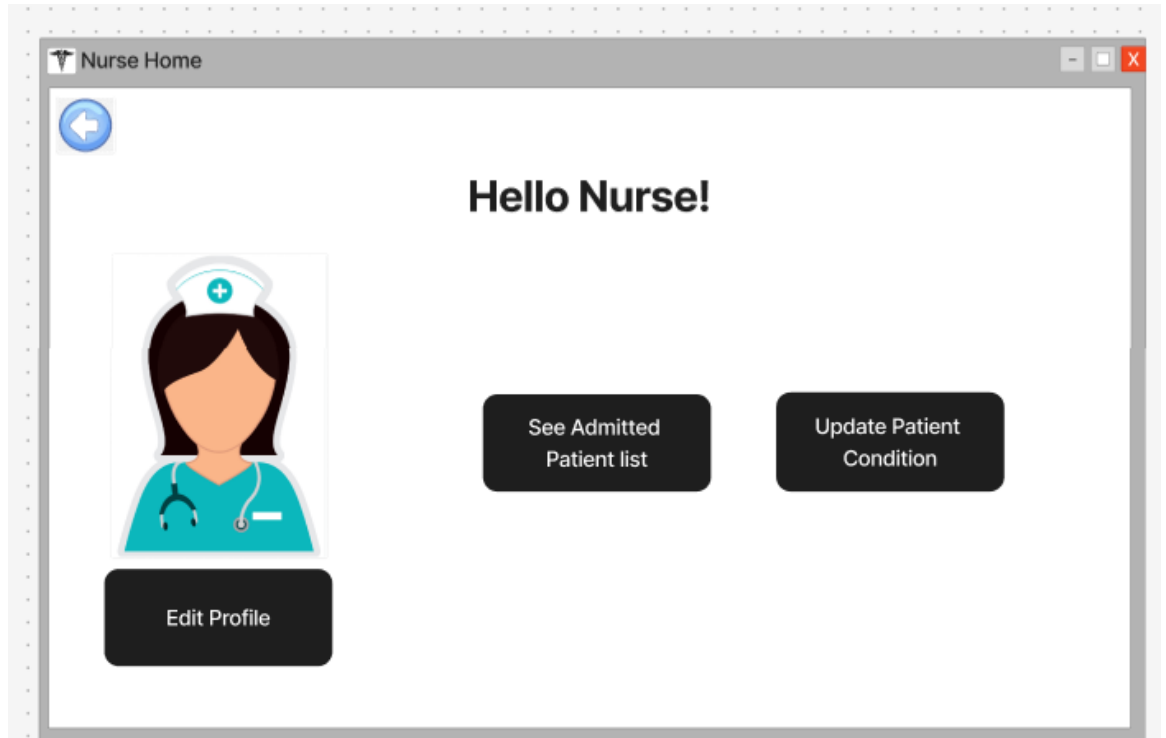
Class Diagram:



Class Diagram:

3.2 UI / Wireframe Design using Figma

Nurse-



A wireframe of a web application window titled "Nurse Signup". The window has a standard macOS-style title bar with minimize, maximize, and close buttons. Inside the window, there is a blue circular button with a white left-pointing arrow in the top-left corner. The main content area features the text "Sign up" in a large, bold, black font. Below this text, there is a form with the following fields and labels: "Name :", "Password :", "Phone :", "Gender :", "Email :", "Qualification :", "Address :", and "Age :". The "Gender :" field has two radio buttons, one labeled "Male" and one labeled "Female". At the bottom of the form is a dark gray button with the text "Create Account" in white.

Nurse Login



Name :

Password :

LOG IN

Don't have an account? [Sing up](#)

Nurse PatientList



Show

Patient name :

Cause :

Recommendation :

Room No :

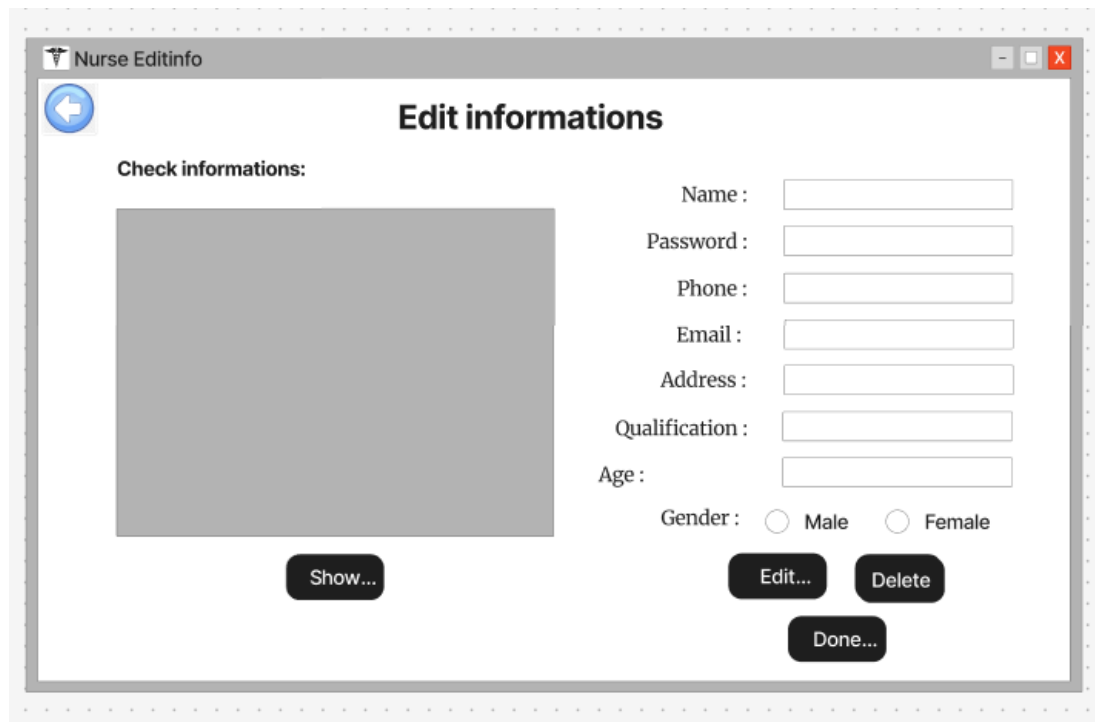
Bed No :

Condition :

ID :

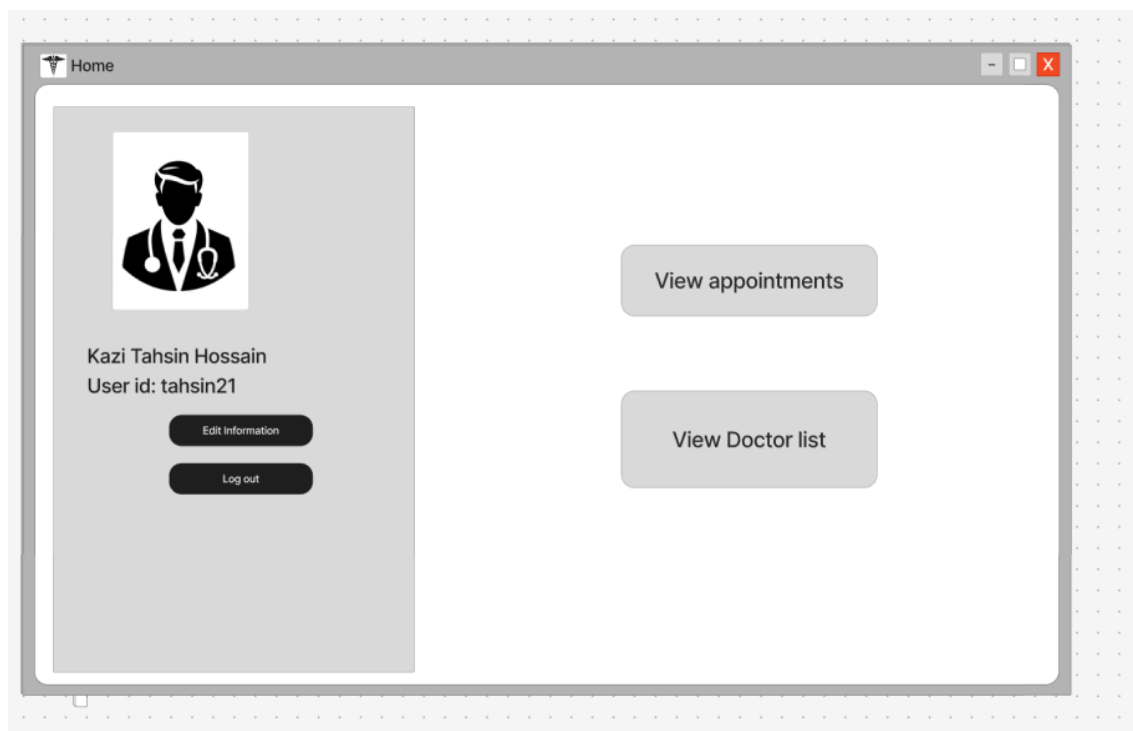
Update

Done



The screenshot shows a web application window titled "Nurse Editinfo". Inside, there's a section titled "Edit informations" with a back arrow icon. Below this, it says "Check informations:" followed by a large grey rectangular placeholder for a profile picture. To the right of the placeholder are several input fields: "Name:", "Password:", "Phone:", "Email:", "Address:", "Qualification:", and "Age:". Below these fields are radio buttons for "Gender:" with options "Male" and "Female". At the bottom right of the form area are three buttons: "Edit...", "Delete", and "Done...". A "Show..." button is located below the profile picture placeholder.

Doctor-



The screenshot shows a web application window titled "Home". On the left side, there's a profile card for a doctor. It features a silhouette icon of a doctor with a stethoscope. Below the icon, the text reads "Kazi Tahsin Hossain" and "User id: tahsin21". Underneath this text are two buttons: "Edit information" and "Log out". On the right side of the window, there are two large, light-grey buttons: "View appointments" and "View Doctor list".

Doctor Signup

Sign up

←

Name :

Password :

Phone :

Email :

Address :

Qualification :

Specialization :

Gender : ☐ Male ☐ Female

Create Account

Doctor Login

Doctor Login

←

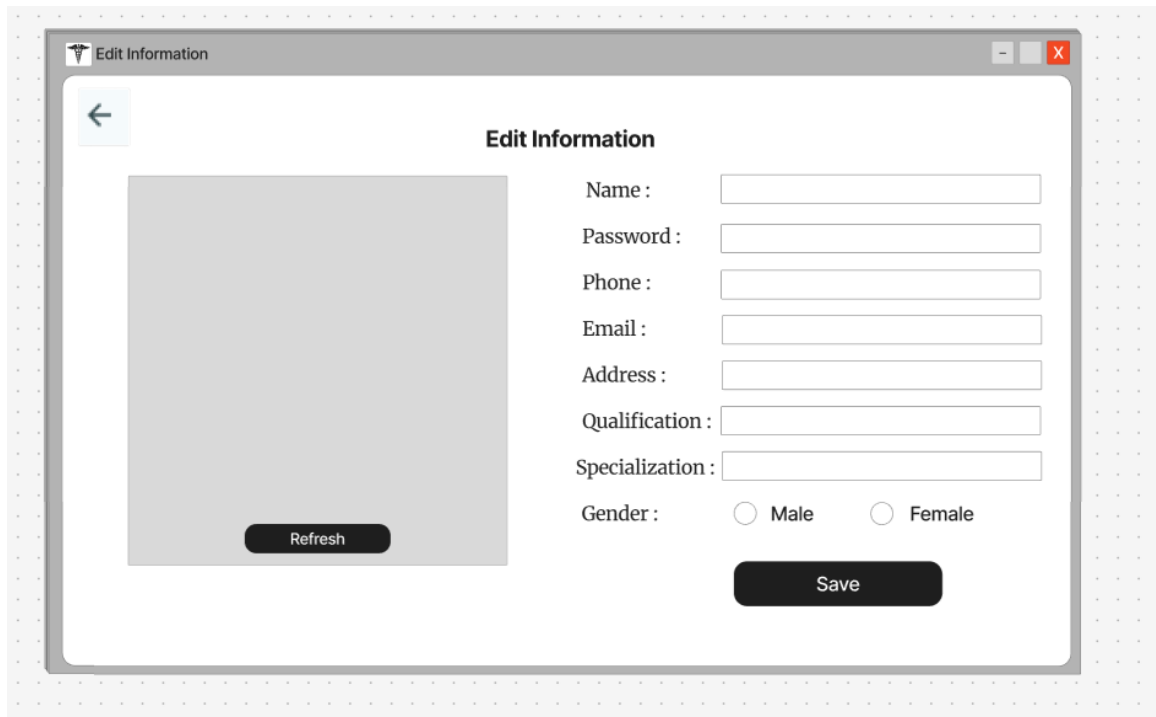


Name :

Password :

LOG IN

Don't have an account? [Sign up](#)



The 'Edit Information' form is a web interface for updating user details. It features a title bar with a back arrow, a main title 'Edit Information', and a large gray placeholder for a profile picture with a 'Refresh' button. To the right, there are input fields for Name, Password, Phone, Email, Address, Qualification, and Specialization. A 'Gender' section includes radio buttons for 'Male' and 'Female'. A 'Save' button is positioned at the bottom right.

Edit Information

←

Refresh

Name :

Password :

Phone :

Email :

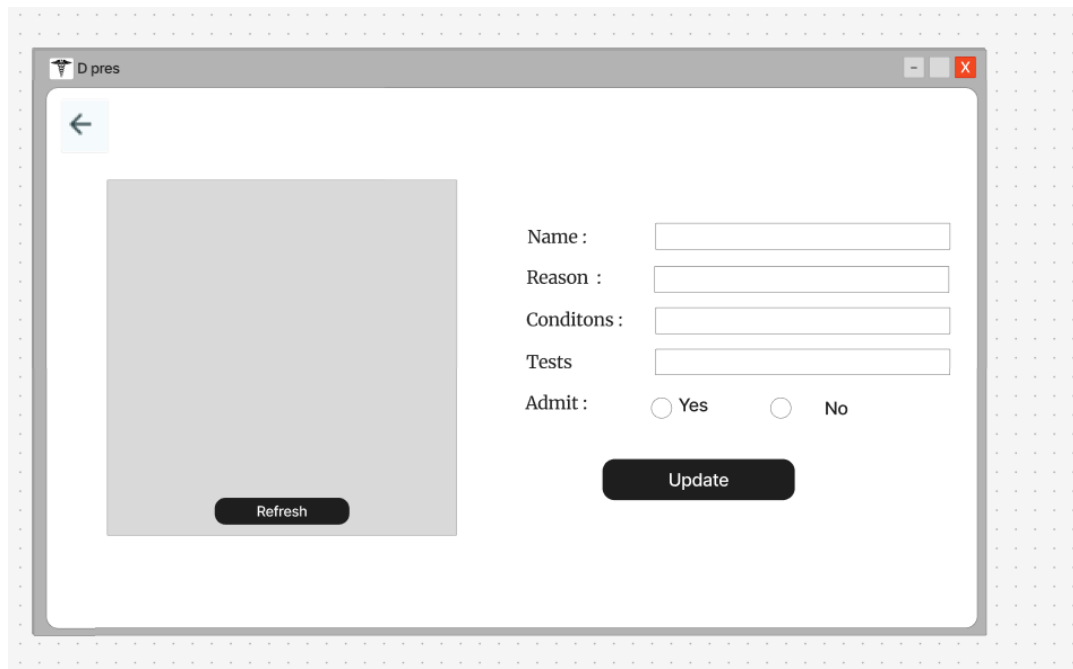
Address :

Qualification :

Specialization :

Gender : ☐ Male ☐ Female

Save



The 'D pres' form is a web interface for updating medical or patient information. It features a title bar with a back arrow, a main title 'D pres', and a large gray placeholder for a profile picture with a 'Refresh' button. To the right, there are input fields for Name, Reason, Conditions, and Tests. A section for 'Admit' includes radio buttons for 'Yes' and 'No'. An 'Update' button is positioned at the bottom right.

D pres

←

Refresh

Name :

Reason :

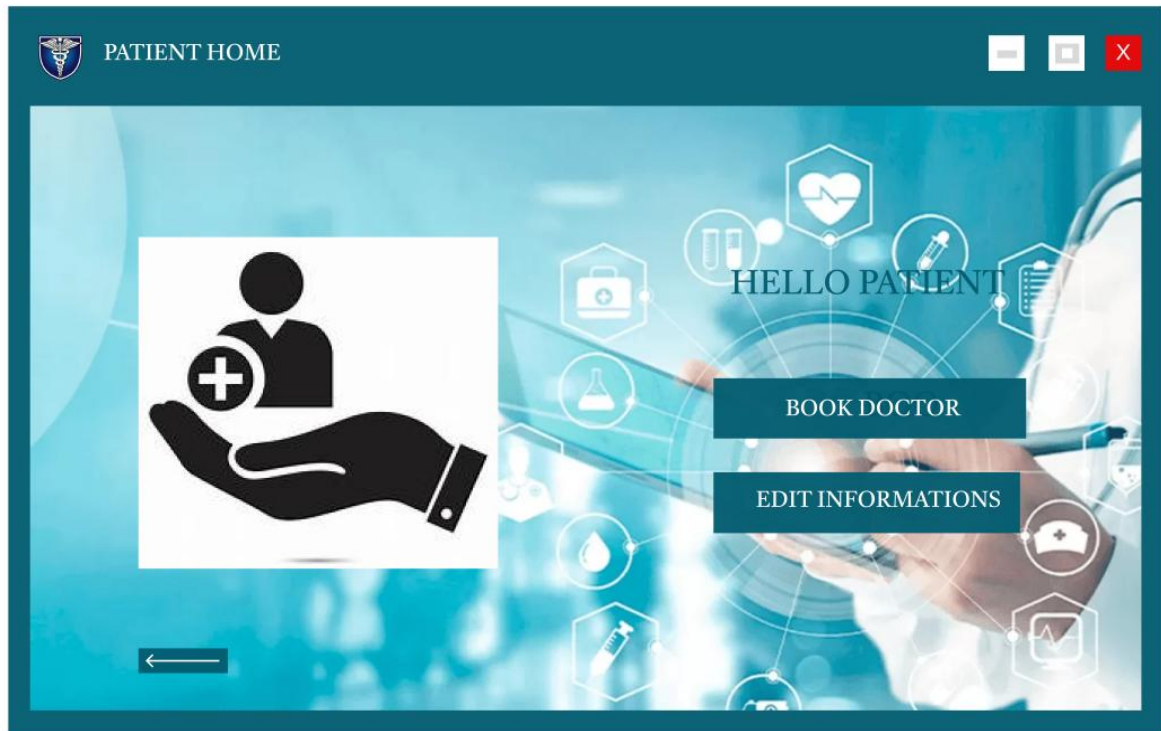
Conditions :

Tests :

Admit : ☐ Yes ☐ No

Update

Patient-



The screenshot shows a web application window titled "PATIENT'S SIGN UP". The interface features a dark teal header with a medical logo on the left and window control buttons (minimize, maximize, close) on the right. The main content area has a light blue background with a blurred image of a doctor's hand holding a stethoscope. On the left, there is a white square icon depicting a person with a plus sign inside a circle, held by a hand. To the right of this icon, the text "HELLO PATIENT" is displayed in a serif font. Below this text are two dark teal buttons with white text: "BOOK DOCTOR" and "EDIT INFORMATION". The background is decorated with various medical icons in white hexagons, including a heart with a pulse line, a test tube, a flask, a syringe, a pill, and a stethoscope. A small white arrow icon is located at the bottom left of the main content area.

SIGN UP

NAME

PHONE NUMBER

EMAIL

ADDRESS

GENDER ☐ MALE ☐ FEMALE

BLOOD GROUP

AGE

PASSWORD

SIGN IN

 PATIENT'S LOG IN



LOG IN

NAME

PASSWORD

DON'T HAVE AN ACCOUNT?

 BOOK DOCTOR'S

BOOK DOCTOR

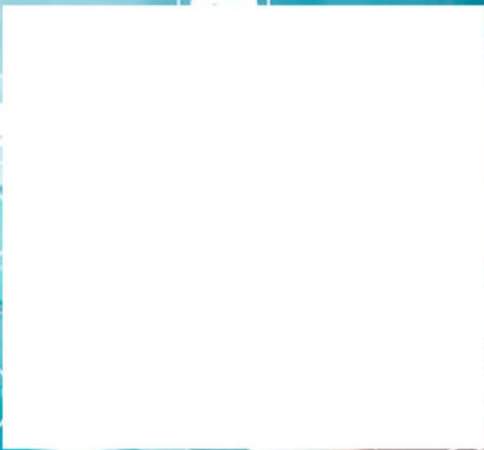
DOCTOR NAME

TIMING

PATIENT NAME

REASON

CONDITION





PATIENT EDIT INFORMATION

EDIT

NAME

PHONE NUMBER

EMAIL

ADDRESS

GENDER ☐ MALE ☐ FEMALE

BLOOD GROUP

AGE

ID

PASSWORD

← EDIT DELETE DONE

SHOW →

Receptionist-

R_BOOK

HELLO!!

NAME SHEDULE

EDIT INFORMATION

BOOK BED

SIGNUP.R

SIGN UP

NAME

PASSWOD

PHONE

GENDER ☐ MALE ☐ FEMALE

EMAIL

QUALIFICATION

EXPERIANCE

AGR

SIGN UP

LOGIN.R

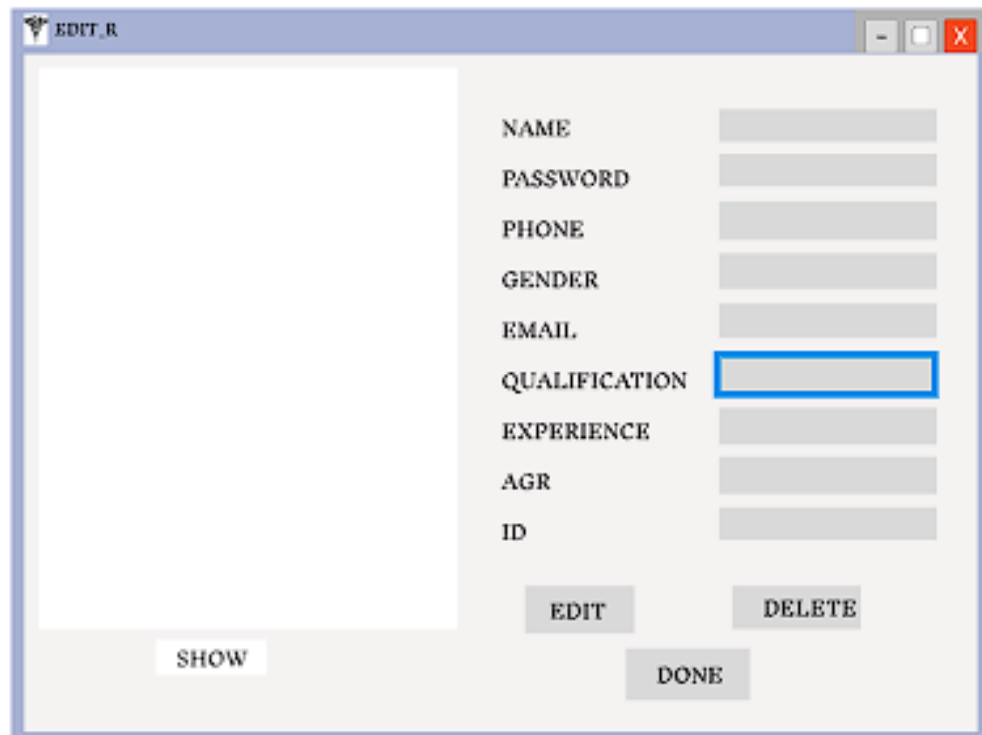


NAME

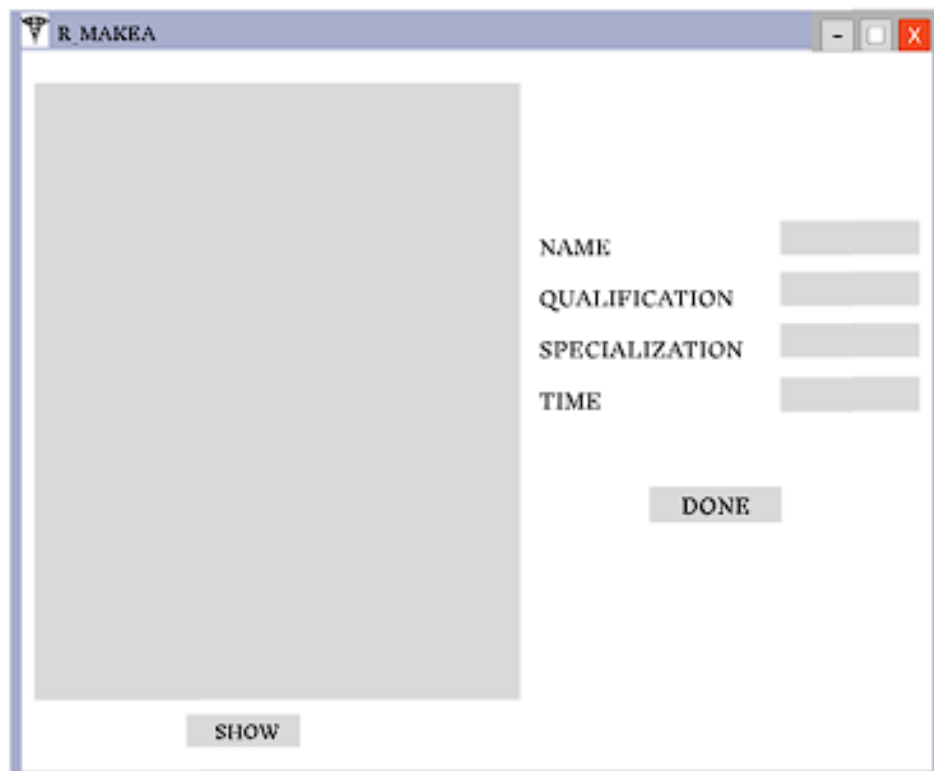
PASSWORD

LOG IN

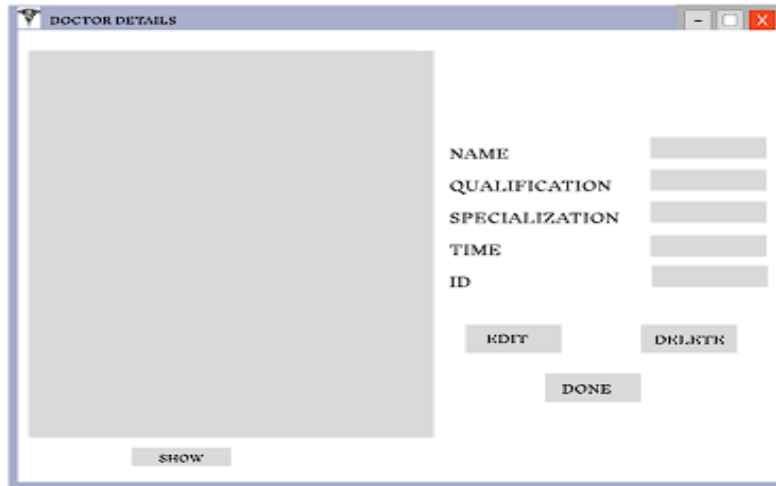
DON'T HAVE ANY ACCOUNT **SING UP**



The screenshot shows a window titled "EDIT_R" with a standard Windows-style title bar. The window is divided into two main sections. On the left is a large, empty white rectangular area. On the right is a form with several text input fields, each preceded by a label: "NAME", "PASSWORD", "PHONE", "GENDER", "EMAIL", "QUALIFICATION", "EXPERIENCE", "AGR", and "ID". The "QUALIFICATION" field is highlighted with a blue rectangular border. Below the form, there are three buttons: "EDIT" and "DELETE" are positioned side-by-side, and "DONE" is centered below them. At the bottom left of the window, there is a "SHOW" button.

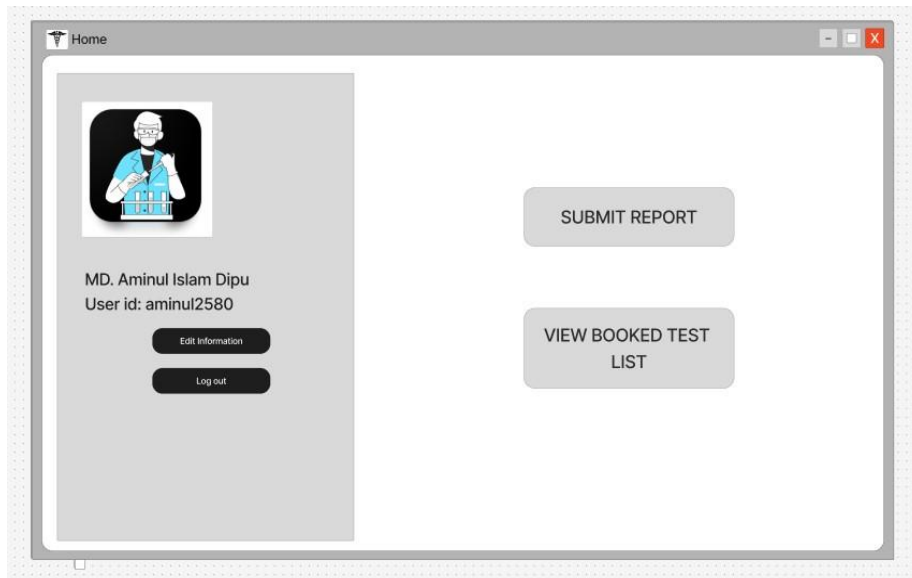


The screenshot shows a window titled "R_MAKEA" with a standard Windows-style title bar. The window is divided into two main sections. On the left is a large, empty light gray rectangular area. On the right is a form with four text input fields, each preceded by a label: "NAME", "QUALIFICATION", "SPECIALIZATION", and "TIME". Below the form, there is a single "DONE" button. At the bottom left of the window, there is a "SHOW" button.

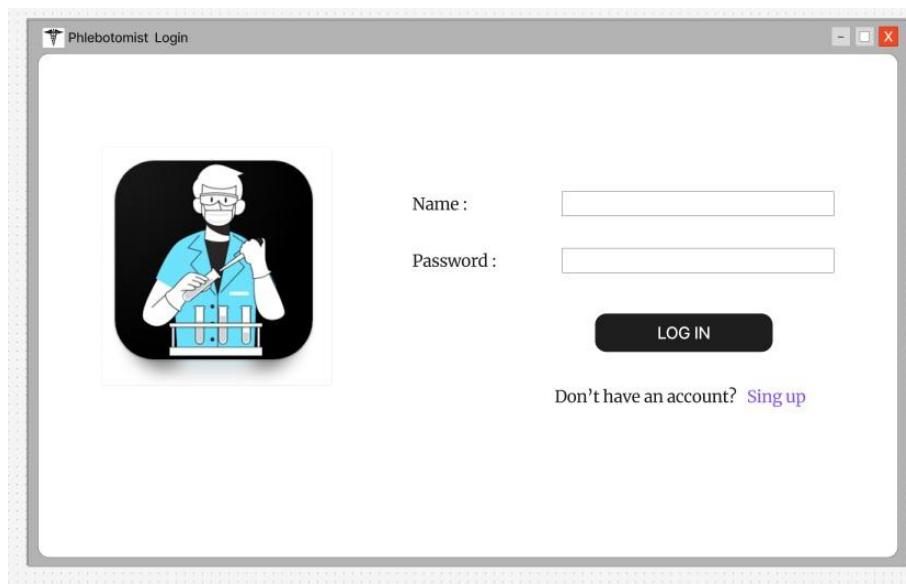


A screenshot of a web application window titled "DOCTOR DETAILS". The window has a large gray rectangular area on the left, likely for a profile picture, with a "SHOW" button below it. On the right, there are five text input fields labeled "NAME", "QUALIFICATION", "SPECIALIZATION", "TIME", and "ID". Below these fields are three buttons: "EDIT", "DELETE", and "DONE".

Phlebotomist-



A screenshot of a web application window titled "Home". The window displays a user profile on the left with a cartoon illustration of a phlebotomist, the name "MD. Aminul Islam Dipu", and the user ID "User id: aminul2580". Below the profile are two buttons: "Edit Information" and "Log out". On the right side of the window, there are two large buttons: "SUBMIT REPORT" and "VIEW BOOKED TEST LIST".



A screenshot of a web application window titled "Phlebotomist Login". The window features a cartoon illustration of a phlebotomist on the left. To the right, there are two text input fields labeled "Name :" and "Password :". Below these fields is a "LOG IN" button. At the bottom, there is a link that says "Don't have an account? [Sing up](#)".

Phlebotomist Signup

Sign up

←

Name :

Password :

Phone :

Email :

Address :

Qualification :

Department :

Gender : ☐ Male ☐ Female

Create Account

Book Test List

Book Test

Name :

Phone :

Email :

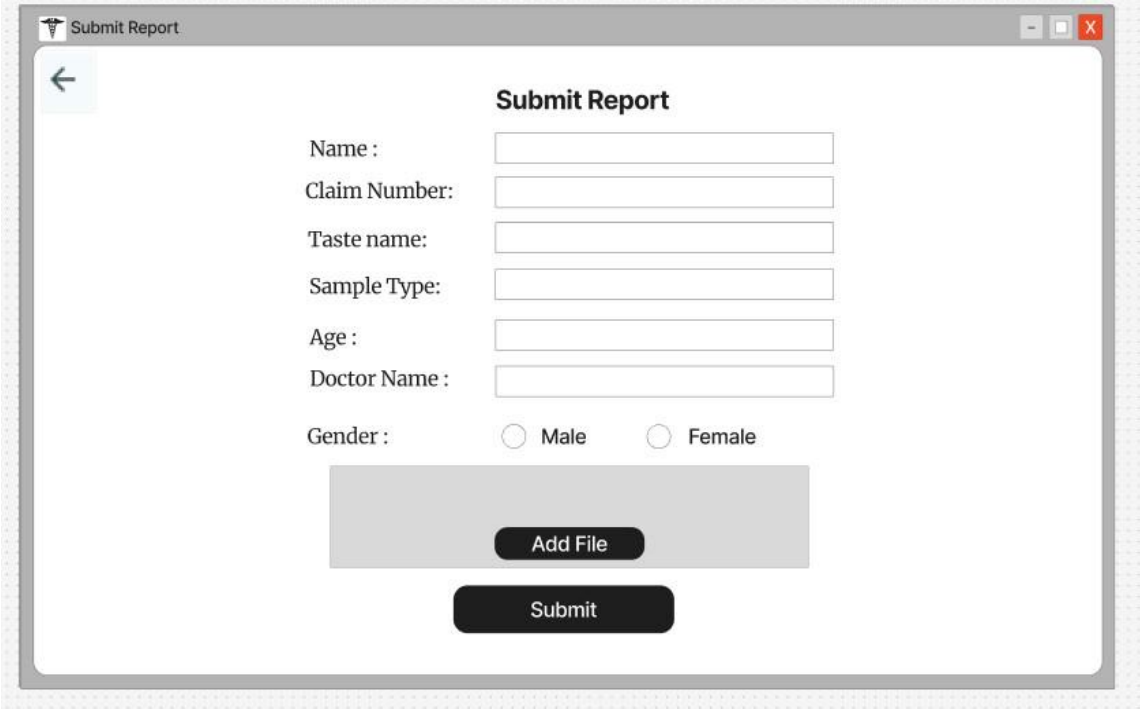
Address :

Test Name :

Doctor Name :

Gender : ☐ Male ☐ Female

Save



Submit Report

←

Submit Report

Name :

Claim Number:

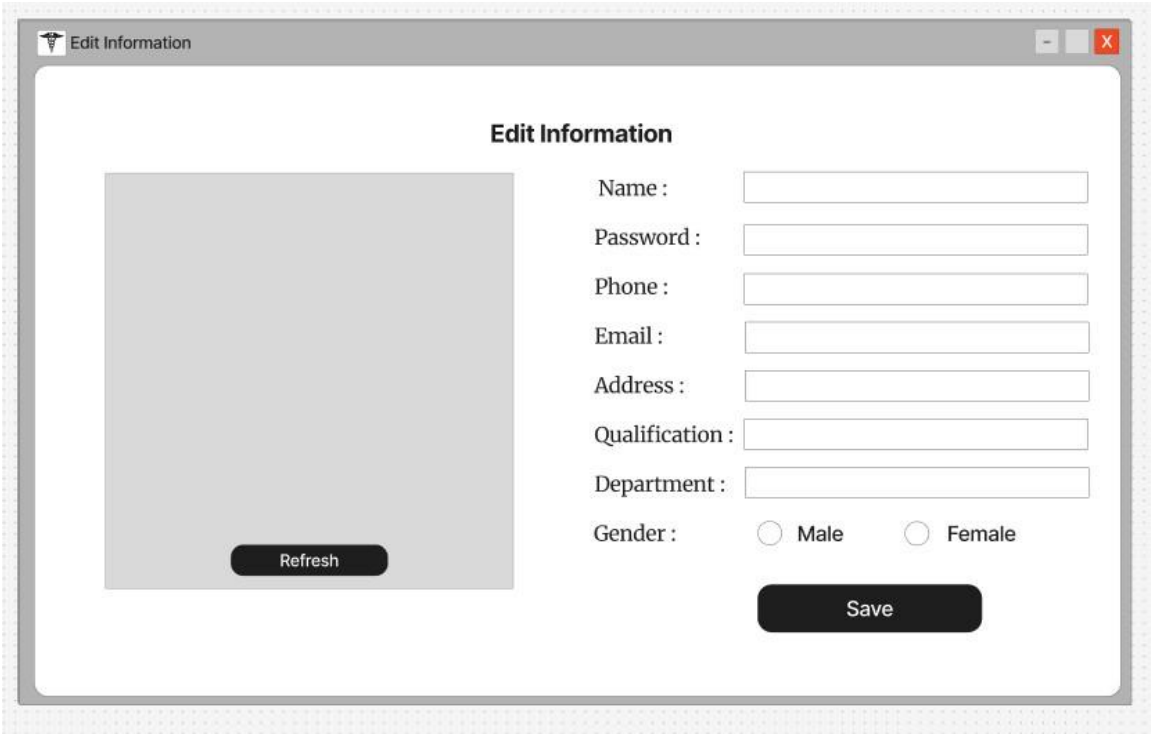
Taste name:

Sample Type:

Age :

Doctor Name :

Gender : ☐ Male ☐ Female



Edit Information

Edit Information

Name :

Password :

Phone :

Email :

Address :

Qualification :

Department :

Gender : ☐ Male ☐ Female

4. GIT WORKFLOW

Commits

 nur ▾

All users ▾ All time ▾

Commits on Jan 20, 2026

final commit

 tanjil-007 committed 12 minutes ago

5034dc1  <>

Commits on Jan 2, 2026

1st file

 tanjil-007 committed 3 weeks ago

e16731f  <>

Commits on Jan 1, 2026

llm.docx

 tanjil-007 committed 3 weeks ago

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Commits

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Commits on Jan 20, 2026

doctor all

 tahsinrupol committed 15 hours ago

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Commits on Jan 20, 2026

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 Touhid committed 15 hours ago

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Commits on Jan 20, 2026

phlebotomist ui and test case

 Aminul2700 committed 18 hours ago

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Commits on Jan 20, 2026

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 ferdousimtiajsraben2878-sr committed 15 hours ago

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5. SOFTWARE TESTING

Receptionist-

Project Name: Hospital management system		Test Designed by: Ferdous Imtiaj Srabon		
Test case Id:TC_1		Test Designed Date: 1.5.2026		
Test priority (easy, medium, high): medium		Test Executed by:Name		
Module name:Edit Session		Test Execution Date:19.01.26		
Test Title:Edit information				
Description:Test the webside edit page				
Precondition:Sucessful login				
Dependencies:				
Test Step	Test Data	Expected result	Actual result	Status(fail/pass)
1.Go to the site 2.Sucessful login 3.Click the Edit information button 4.Edit as required 5.Click Edit button	Name: rrre55 Password:555	The user should edit and delete application	As expected	pass

Project Name: Hospital management system		Test Designed by: Ferdous Imtiaj Srabon		
Test case Id:TC_2		Test Designed Date: 1.5.2026		
Test priority (easy, medium, high): medium		Test Executed by:		
Module name:Home page		Test Execution Date:		
Test Title:Verify home page Navigation Buttons Functionality				
Description:Verify the home page				
Precondition:user has access to the system				
Dependencies: Schedule management module, information edit and brd booking module				
Test Step	Test Data	Expected result	Actual result	Status(fail/pass)
1.Launch the application 2. Welcome message 3. Edit schedule 4.Edit information 5.Book bed	Hello	Home page loads successfully Welcome message display Redirects to edit information page Redirects to Bed Booking page	As Expected, all the button	pass

Project Name: Hospital management system		Test Designed by: Ferdous Intiaj Srabon		
Test case Id:		Test Designed Date: 1.5.2026		
Test priority (easy, medium, high): medium		Test Executed by:		
Module name: Sign Up		Test Execution Date:		
Test Title:verify user sign-up with valid details				
Description: This test case verifies that a new user can successfully register by providing valid information such as name, password, phone number, gender, email, qualification, experience, and age.				
Precondition: User is on the Sign-Up page				
Dependencies: User Authentication Module				
Test Step	Test Data	Expected result	Actual result	Status(fail/pass)
1Open Sign Up page 2. Enter Name 3.Enter Password 4.Enter Phone Number 5. Select Gender 6. Enter Email 7.Enter Qualification 8. Enter Experience 9. Enter Age 10. Click Sign Up button	Name:eeee Password:638 Phone:393939 Gender:male Email:sss@.com Qualification:BSc Experience:phd Age:55	All the information successfully valid and then accept	As expected,	pass

Project Name: Hospital management system		Test Designed by: Ferdous Imtiaj Srabon		
Test case Id:		Test Designed Date: 1.5.2026		
Test priority (easy, medium, high): medium		Test Executed by:		
Module name: Doctor List Management		Test Execution Date:		
Test Title: Verify system behavior when editing doctor details with invalid or missing data				
Description: This test case verifies that the system does not allow editing or saving doctor information when required fields (name, qualification, time, ID) are missing or invalid, and proper error messages are shown.				
Precondition: Doctor list page is opened				
Dependencies: Doctor Management Module				
Test Step	Test Data	Expected result	Actual result	Status(fail/pass)
1. Open Doctor List page 2. Click Edit button 3. Clear Doctor Name field 4. Enter invalid Qualification 5. Leave Time field empty 6. Enter invalid Doctor ID 7. Click Done 8. Click Delete without confirmation	Doctor ID: D101 #### s@	Doctor list displayed	Error	Fail

Patient-

Project Name: Hospital management system		Test Designed by: Touhid A Akbar		
Test case Id:TC_1		Test Designed Date: 1.19.2026		
Test priority (easy, medium, high): medium		Test Executed by:		
Module name: Patient signup functionality		Test Execution Date: 1.19.2026		
Test Title: Verify patient signup				
Description: Test the patient module to verify that a patient can sign up				
Precondition: The patient must be registered in the system and the application must be running properly.				
Dependencies: Database connection, Doctor module, Patient module				
Test Step	Test Data	Expected result	Actual result	Status(fail/pass)
1. Open the application 2. Click Sign Up 3. Enter patient details 4. Click SIGNIN	Name: Touhid Phone: 017xxxxxxx Email: touhid@gmail.com Address: Dhamrai Gender: Male Blood Group: O+ Age:24 Password: 12345	Patient account should be created successfully	As expected	Pass

Project Name: Hospital management system		Test Designed by: Touhid A Akbar		
Test case Id:TC_2		Test Designed Date: 1.19.2026		
Test priority (easy, medium, high): medium		Test Executed by:		
Module name: Booking		Test Execution Date:		
Test Title: Doctor booking				
Description: Test the patient module to verify that a patient can book a doctor				
Precondition: The patient must be registered in the system and the application must be running properly.				
Dependencies: Database connection, Doctor module, Patient module				
Test Step	Test Data	Expected result	Actual result	Status(fail/pass)
1. Click Book Doctor 2. Select doctor name 3. Select time 4. Enter patient name 5. Enter reason 6. Enter condition 7click BOOK	Doctor: Dr. Rahman Time: 10 AM Patient name: TOUHID Reason:advice Condition:Bad	Booking form should open successfully	As expected	PASS

Project Name: Hospital management system		Test Designed by: Touhid A Akbar		
Test case Id:TC-3		Test Designed Date: 1.19.2026		
Test priority (easy, medium, high): medium		Test Executed by:		
Module name: Booking		Test Execution Date: 1.19.2026		
Test Title: Doctor booking				
Description: Book a doctor				
Precondition: The patient must be registered in the system and the application must be running properly.				
Dependencies: Database connection, Doctor module, Patient module				
Test Step	Test Data	Expected result	Actual result	Status(fail/pass)
1. Enter booking reason 2. Click Book	Reason: Fever	Appointment should be booked successfully	Booking failed	Fail

Project Name: Hospital management system		Test Designed by: Touhid A Akbar		
Test case Id:TC-4		Test Designed Date: 1.5.2026		
Test priority (easy, medium, high): medium		Test Executed by:		
Module name: Edit Information		Test Execution Date:		
Test Title: Edit information functionality				
Description: Test the patient module to verify that a patient can edit personal information successfully.				
Precondition: The patient must be registered in the system and the application must be running properly.				
Dependencies: Database connection, Doctor module, Patient module				
Test Step	Test Data	Expected result	Actual result	Status(fail/pass)
1. Click Edit Information 2. Update phone number 3. Click DONE	Phone: 01867890901	Patient information should be updated	As expected	Pass

Doctor-

Project Name: Hospital management system		Test Designed by: Kazi Tahsin Hossain		
Test case Id:TC-91		Test Designed Date: 1.5.2026		
Test priority (easy, medium, high): medium		Test Executed by:		
Module name: Edit Information		Test Execution Date:		
Test Title: Edit information functionality				
Description: Test the doctor module to verify that a doctor can edit doctor information successfully.				
Precondition: The doctor must be registered in the system and the application must be running properly.				
Dependencies: Database connection, Doctor module.				
Test Step	Test Data	Expected result	Actual result	Status(fail/pass)
1. Click Edit Information 2. Update email 3. Click DONE	Email: tahsin@gmail.com	Doctor information should be updated	As expected	Pass

Project Name: Hospital management system		Test Designed by: Kazi Tahsin hossain		
Test case Id:TC_90		Test Designed Date: 1.19.2026		
Test priority (easy, medium, high): medium		Test Executed by: rupol		
Module name: doctor signup information functionality		Test Execution Date: 1.19.2026		
Test Title: Verify doctor signup page				
Description: Test the doctor module to verify that a doctor can sign up				
Precondition: The doctor must be registered in the system and the application must be running properly.				
Dependencies: Database connection, Doctor module.				
Test Step	Test Data	Expected result	Actual result	Status(fail/pass)
1. Open the application 2. Click Sign Up 3. Enter doctor details 4. Click SIGNIN	Name: tahsin Phone: 017xxxxxxx Email: tah@gmail.com Address: dhaka Gender: Male Blood Group: O+ Age:20 Password: 12345	doctor signup should be created successfully	As expected	Pass

Project Name: Hospital management system		Test Designed by: Kazi Tahsin Hossain		
Test case Id:TC92		Test Designed Date: 1.19.2026		
Test priority (easy, medium, high): medium		Test Executed by:Tester		
Module name: Doctor Homepage		Test Execution Date: 1.19.26		
Test Title: Verify Doctor Home page Navigation				
Description: To verify that doctor can view appointments and edit info from homepage.				
Precondition: Doctor must be registered and successful login.				
Dependencies: login module, database connection				
Test Step	Test Data	Expected result	Actual result	Status(fail/pass)
1.Verify welcome text. 2.click on “view appointment” button 3.click on “edit information ”buttn	1.welcome text	1.appointment list should open 2.edit information page should open	As expected	Pass

Project Name: Hospital management system		Test Designed by: Kazi Tahsin Hosssain		
Test case Id:TC93		Test Designed Date: 1.19.2026		
Test priority (easy, medium, high): medium		Test Executed by:Tester		
Module name:Doctor Module		Test Execution Date:1.19.26		
Test Title:Verify doctor can search and update patient information.				
Description:To verify that doctor can search patient details and update condition				
Precondition: successful login, patient record must exist in database				
Dependencies: appointment module, patient modeule, database connectivity				
Test Step	Test Data	Expected result	Actual result	Status(fail/pass)
1.enter patient id 2.click search button 3.update condition field 4.click update button 5.click done button	1.valid patient id 2.update condition text 3.update data	1.patient details should load. 2.patient data should be displayed. 3.condition should update 4.database should save the data.	The condition update is not working .	Failed .

Phlebotomist-

Project Name: Hospital management system		Test Designed by: Md. Aminul Islam Dipu		
Test case Id: Test Case 1		Test Designed Date: 19.01.2026		
Test priority (easy, medium, high): medium		Test Executed by: Md. Aminul Islam Dipu		
Module name: Phlebotomist signup functionality		Test Execution Date: 19.01.2026		
Test Title: Verify phlebotomist signup				
Description: Test the phlebotomist module to verify that a phlebotomist can sign up.				
Precondition: The phlebotomist must be registered in the system and the application must be running properly.				
Dependencies: Database connection, Doctor module, Patient module.				
Test Step	Test Data	Expected result	Actual result	Status(fail/pass)
1. Open the application 2. Click Sign Up 3. Enter phlebotomist details 4. Click Create Account	Name: Aminul Password: 12345 Phone:017xxxxxxx Email:aminul@gmail.com Address: Mirpur-12 Qualification: Bsc Department: Pathology Gender: Male	Phlebotomist account should be created successfully	As expected	Pass

Project Name: Hospital management system		Test Designed by: Md. Aminul Islam Dipu		
Test case Id: Test Case 2		Test Designed Date: 19.01.2026		
Test priority (easy, medium, high): medium		Test Executed by: Md. Aminul Islam Dipu		
Module name: Phlebotomist Login Functionality		Test Execution Date: 19.01.2026		
Test Title: Verify Phlebotomist Login				
Description: Test the Phlebotomist login module to verify that a registered phlebotomist can log in successfully.				
Precondition: The phlebotomist must already be registered in the system. Application must be running.				
Dependencies: Database connection, valid phlebotomist credentials, signup module completed.				
Test Step	Test Data	Expected result	Actual result	Status(fail/pass)
1. Open the application 2. Enter Name 3. Enter Password 4. Click LOG IN	Name: Aminul Password: 12345	Login successful, redirected to Home	As expected	Pass

Project Name: Hospital management system		Test Designed by: Md. Aminul Islam Dipu		
Test case Id: Test Case 3		Test Designed Date: 19.01.2026		
Test priority (easy, medium, high): medium		Test Executed by: Md. Aminul Islam Dipu		
Module name: Book Test Functionality		Test Execution Date: 19.01.2026		
Test Title: Verify Book Test Form Submission				
Description: Test the Book Test form to verify that a phlebotomist can book a test for a patient.				
Precondition: Phlebotomist must be logged in. Patient and doctor data should be available.				
Dependencies: Login module, patient and doctor modules, database connectivity.				
Test Step	Test Data	Expected result	Actual result	Status(fail/pass)
1. Navigate to Book Test page	Name: John Doe	Data saved,	As expected	Pass
2. Fill Name field	Phone: 017xxxxxxxxx	confirmation		
3. Fill Phone field	Email:john@test.com	shown		
4. Fill Email field	Address: Dhaka			
5. Fill Address field	Test Name: CBC Doctor Name: Dr.			
6. Fill Test Name field	Smith			
7. Fill Doctor Name field	Gender: Male			
8. Select Gender				
9. Click Save				

Project Name: Hospital management system		Test Designed by: Md. Aminul Islam Dipu		
Test case Id: Test Case 4		Test Designed Date: 19.01.2026		
Test priority (easy, medium, high): medium		Test Executed by: Md. Aminul Islam Dipu		
Module name: Submit Report Functionality		Test Execution Date: 19.01.2026		
Test Title: Verify Submit Report Function				
Description: Test the Submit Report module to verify that a phlebotomist can submit a lab report for a patient.				
Precondition: Phlebotomist must be logged in. A booked test must exist for the patient.				
Dependencies: Login module, Book Test module, file upload functionality, database.				
Test Step	Test Data	Expected result	Actual result	Status(fail/pass)
1.Navigate to Submit Report page 2.Fill Name field 3.Fill Claim Number 4.Fill Taste name (Test Name) 5.Fill Sample Type 6.Fill Age 7.Fill Doctor Name 8.Select Gender 9.Click Add File 10. Click Submit	Name: John Doe Claim No: CL12345 Test Name: CBC Sample Type: Blood Age: 30 Doctor Name: Dr. Smith Gender: Male File: report.pdf	Report submitted successfully	Error: "Invalid Claim Number" displayed. Submission failed.	Fail

Nurse-

Project Name: Hospital Management System.			Test Designed by: Tanjil		
Test Case ID: TN-1			Test Designed date: 19 Jan, 2026		
Test Priority: High			Test Executed by: Srabon		
Module Name: Login Session			Test Executed date: 19 Jan, 2026		
Test Title: Verify Login with valid username and password					
Description: Test the website login Page					
Precondition: The user has a valid username and password					
Dependency: Username and password					
Test Steps		Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to the site 2. Enter information 3. Click submit		Username: nurse1 Password: 1234	The user should log in to the site	As expected	Pass

Project Name: Hospital Management System.		Test Designed by: Tanjil		
Test Case ID: TN-2		Test Designed date: 19 Jan, 2026		
Test Priority: Medium		Test Executed by: Srabon		
Module Name: Signup Session		Test Executed date: 19 Jan, 2026		
Test Title: Verify Signup with valid information				
Description: Test the website signup page for nurse				
Precondition: Nurse must not be registered before				
Dependency:				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
Go to signup page, enter valid details, click submit	Name: nurse1, Email: nurse@gmail.com, Password: 1234	Nurse account should be created successfully	As expected	Pass

Project Name: Hospital Management System.		Test Designed by: Tanjil		
Test Case ID: TN-3		Test Designed date: 19 Jan, 2026		
Test Priority: High		Test Executed by: Srabon		
Module Name: See admitted patient list		Test Executed date: 19 Jan, 2026		
Test Title: Show the admitted patient list				
Description: Test the website patient list				
Precondition: Nurse should be registered in the database				
Dependency: List should be updated				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
Login as nurse, go to patient list page	Username: nurse1, Password: 1234	List of admitted patients should be displayed	As expected	Pass

Project Name: Hospital Management System.		Test Designed by: Tanjil		
Test Case ID: TN-4		Test Designed date: 19 Jan, 2026		
Test Priority: Low		Test Executed by: Srabon		
Module Name: Update patient list		Test Executed date: 19 Jan, 2026		
Test Title: Update patient condition				
Description: Test the update condition feature for nurse				
Precondition: Nurse must be logged in and patient must be admitted				
Dependency:				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
Login as nurse, go to patient list, select patient, update condition	Patient ID: 101, Condition: Stable	Patient condition should be updated	Condition not updated due to error	Fail

6. CONCLUSION

The Hospital Management System (HMS) is designed to improve the efficiency and accuracy of hospital operations by providing a centralized platform for all users, including doctors, nurses, receptionists, phlebotomists, and patients. Traditional hospital management methods often involve manual record-keeping and fragmented communication, which can cause delays, errors, and difficulties in tracking patient information. This system addresses these problems by integrating various functions such as appointment booking, patient admission, medical record management, test assignment, and report submission into a single, user-friendly platform.

Using the Scrum model for development allowed the project team to work iteratively and deliver the system in incremental steps. Each sprint focused on specific functionalities, enabling continuous feedback and improvement. This approach ensured that all requirements, both functional and non-functional, were implemented effectively while maintaining quality and usability. The role-based access system ensures that users can perform their tasks securely and efficiently according to their responsibilities.

Overall, the Hospital Management System reduces administrative workload, enhances patient care, and ensures seamless coordination among hospital staff. It is a scalable and flexible solution that can accommodate future enhancements, making it suitable for modern hospital environments. This project successfully demonstrates practical application of software engineering principles and provides a reliable solution for real-world healthcare management.